GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

3. A scale copy is made at k = 1.5. A side of the original is 6 cm. Find the matching side of the copy.

4. A drawing has a 65° angle. Its scale copy is made at k = 2. What is the matching angle in the copy?

- ○ A. 67°
- ○ B. 130°
- ○ C. 32.5°
- ○ D. 65°

5. A map is drawn at 1 cm : 30 km. Two towns are 4.5 cm apart on it. How far apart are they?

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · 6 questions

6. A figure with perimeter 16 cm is scaled by k = 2.5. Find the copy's perimeter.

7. A figure with area 9 cm^2 is scaled by k = 2.5. Find the copy's area.

8. A 4 cm by 6 cm rectangle is scaled by k = ½.

a. Find the copy's perimeter.

b. Find the copy's area.

9. A figure has area 10 cm^2 and its scale copy has area 160 cm^2. Find k.

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

PART C. Real-World Scale Drawings

Lesson 4 · 4 questions

12. A plan is drawn at ½ in : 1 ft. A wall is 7.5 in long on it. Find the real length.

13. A plan is drawn at 1 in : 6 ft. A real wall is 21 ft long. Find its length on the plan.

14. A model boat is built at 1 : 25 and is 32 cm long. Find the real boat's length in metres.

15. Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m.

a. Find the real length of the wall drawn 3 in long.

b. Find the room's real area.

c. Write the scale factor from the drawing to the room as a single number.

d. Find the room's real area in square metres.

PART D. Converting Length Across Systems; Currency

Lessons 6–7 · 6 questions

16. 1 in = 2.54 cm. How many centimetres is 14 in?

17. 1 cm = 0.39 in. How many inches is 120 cm?

18. 1 km = 0.62 mi. How many miles is 30 km?

19. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 6 yd? Estimate first.

20. Rate on 3 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $150?

b. How many US dollars is 69 euros?

21. 1 in = 2.54 cm; 1 cm = 0.39 in. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

PART E. Converting Mass, Area and Volume

Lessons 8–9 · 5 questions

22. 1 lb = 0.454 kg. How many kilograms is 80 lb?

23. 1 L = 0.264 gal. How many gallons is 40 L?

24. 1 ft = 12 in. How many square inches is 8 ft^2?

25. 1 c = 8 fl oz and 1 c = 0.24 L. How many litres is 48 fl oz?

26. 1 m = 100 cm. A desk is 1.5 m^2 in area. How many square centimetres is that?

- ○ A. 300 cm^2
- ○ B. 150 cm^2
- ○ C. 15,000 cm^2
- ○ D. 1,500 cm^2

PART F. Multi-Step Proportion Problems

Lessons 10–11 · 4 questions

27. A car does 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

28. A recipe for 3 servings uses 1¼ cups of rice.

a. How much rice for 12 servings?

b. With 10 cups on hand, how many servings can be made?

29. 1 km = 0.62 mi. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

30. Paint is mixed 1 : 2 : 5. A batch uses 10 gal of the third colour. How many gallons is the whole batch?